

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – X
PAPER –1
TEST DATE: 21-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. D

Sol. $E = Bv$

$$q = \frac{\epsilon_0 A}{d} Ed$$

$$q = \epsilon_0 \pi R^2 Bv$$

$$I = \frac{dq}{dt} = \epsilon_0 \pi R^2 B \frac{dv}{dt}$$

$$m \frac{dv}{dt} = mg - BId$$

$$m \frac{dv}{dt} = mg - \epsilon_0 \pi R^2 B^2 d \frac{dv}{dt}$$

$$\frac{dv}{dt} (m + \epsilon_0 \pi R^2 B^2 d) = mg$$

$$a = \frac{dv}{dt} = \frac{mg}{(m + \epsilon_0 \pi R^2 B^2 d)}$$

2. A

Sol. When the nail B is removed

$$3mg \frac{\ell}{2} = I_A \alpha = \frac{3m\ell^2}{2} \alpha$$

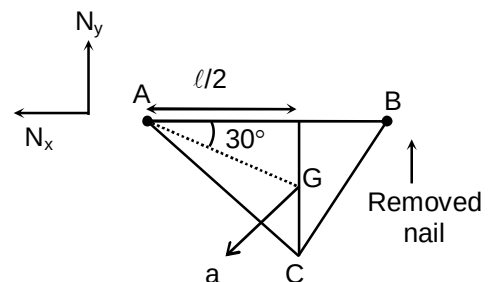
$$\Rightarrow \alpha = g/\ell \quad \dots(i)$$

$$\text{Also, } a = \alpha \frac{\ell}{\sqrt{3}} = \frac{g}{\ell} \times \frac{\ell}{\sqrt{3}} = \frac{g}{\sqrt{3}}$$

$$N_x = 3ma \cos 60^\circ$$

$$\Rightarrow N_x = \frac{\sqrt{3}}{2} mg$$

$$3mg - N_y = 3ma \sin 60^\circ$$



$$\Rightarrow N_y = \frac{3mg}{2}$$

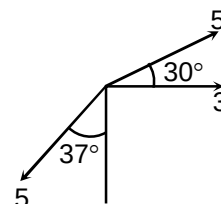
$$N = \sqrt{N_x^2 + N_y^2} = \sqrt{3}mg$$

3. C

 Sol. $V_1 = 3\sin \omega t$; $V_2 = 5\sin(\omega t + \phi_1)$ and $V_3 = 5\sin(\omega t - \phi_2)$

$$V_{\max} = \sqrt{\left(\frac{5\sqrt{3}}{2}\right)^2 + (1.5)^2} = \sqrt{21}$$

$$I_{\max} = \frac{V_{\max}}{R} = \frac{\sqrt{21}}{\sqrt{7/3}} = \sqrt{\frac{21 \times 3}{7}} = 3A$$



4. B

 Sol. $C_{eq} = \frac{C}{2}$ and $C'_{eq} = \frac{kC}{k+1} \Rightarrow C'_{eq} > C_{eq}$
 $\therefore Q' > Q$

5. A

 Sol. $\frac{KA(\theta_1 - \theta_2)}{\ell} = i$... (i)

Thermodynamics process in both the vessels are isochoric

$$-\frac{d\theta_1}{dt} = \frac{i}{2 \times \frac{5R}{2}} = \frac{i}{5R}$$
 ... (ii)

$$\frac{d\theta_2}{dt} = \frac{i}{4 \times \frac{3R}{2}} = \frac{i}{6R}$$
 ... (iii)

Adding (ii) and (iii), we get

$$-\frac{d(\theta_1 - \theta_2)}{dt} = \frac{11i}{30R}$$

$$\Rightarrow -\frac{d(\theta_1 - \theta_2)}{dt} = \frac{11KA(\theta_1 - \theta_2)}{30R\ell}$$

$$-\int_{50}^{25} \frac{d\theta}{\theta} = \frac{kA \times 11}{30\ell R} \int_0^t dt$$
 (where, $\theta_1 - \theta_2 = \theta$)

$$t = \frac{0.693 \times 30 \times 242 \times 10^{-2} \times 8.3 \times 7}{693 \times 7 \times 22 \times 8.3 \times 10^{-4} \times 11} = 3 \text{ sec}$$

6. A, D

 Sol. $E_x = \frac{12}{4} \Rightarrow E_x = 3 \text{ V/m}$

$$E_y = \frac{-8}{2} \Rightarrow E_y = -4 \text{ V/m}$$

7. A, D

Sol. $2T\ell = \rho d\ell hg$

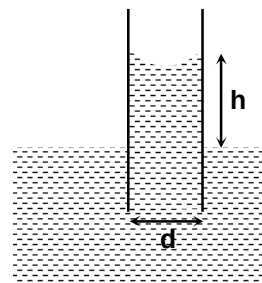
$$h = \frac{2T}{\rho g d} = \frac{2 \times 7 \times 10^{-2}}{10^3 \times 10 \times 1 \times 10^{-4}} = 14 \times 10^{-2} \text{ m} = 14 \text{ cm}$$

Force of mutual attraction between the plates is

$$F = \int_0^h \rho g y \ell dy = \frac{\rho g \ell h^2}{2}$$

$$= \frac{10^3 \times 10 \times 0.10 \times (14 \times 10^{-2})^2}{2}$$

$$= 19.6/2 = 9.8 \text{ N}$$

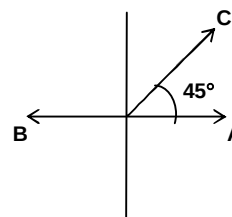


8. B, D

$$\text{Sol. } \sqrt{D^2 + 4d^2} - D = \frac{\lambda}{2}$$

$$\Rightarrow D \left[1 + \frac{1}{2} \times \frac{4d^2}{D^2} - 1 \right] = \frac{\lambda}{2}$$

$$\lambda = \frac{4d^2}{D}$$

Resulting intensity at point P is $I = I_0$ 

9. A, B

$$\text{Sol. } v \propto \frac{1}{n}, r \propto n^2; E \propto \frac{1}{n^2}$$

10. A, C

$$\text{Sol. } F = -\frac{dU}{dr} = -\frac{d}{dr}(k\ell nr) = -\frac{k}{r}$$

$$\frac{mv^2}{r} = \frac{k}{r}; mvr = \frac{nh}{2\pi}; v = \sqrt{\frac{k}{m}}$$

$$\Rightarrow m\sqrt{\frac{k}{m}}r = \frac{nh}{2\pi}; r = \frac{nh}{2\pi\sqrt{mk}}$$

$$\text{Again: } E = U + k = k\ell nr + \frac{1}{2}m\frac{k}{m} = \frac{k}{2} + k\ell nr$$

$$E = k \left[\frac{1}{2} + \ell n \left(\frac{nh}{2\pi\sqrt{mk}} \right) \right]$$

Section – B

11. 40

Sol. Potential across each capacitor is 10 V

Charge = 20 μC Charge on plate C = 40 μC (both sides included)

12. 4

$$\text{Sol. } a_E = \frac{qE_0}{m} \text{ (along negative z-direction)}$$

Electric field will make z-component of velocity zero. At that time speed of the particle will be minimum and that minimum speed is the other component, i.e. v_0 .

This is minimum when,

$$v_z = u_z + a_z t$$

$$\text{Or } 0 = v_0 - \frac{qE_0}{m} t$$

$$\text{Or } t = \frac{mv_0}{qE_0} = 4 \text{ sec}$$

13. 90

Sol. The maximum displacement on each side decreases by $\frac{2\mu mg}{k} = \frac{2 \times 0.4 \times 1 \times 10}{100} = 8 \text{ cm}$

It stops completely if comes to rest between $\pm \frac{\mu mg}{k} = \pm 4 \text{ cm}$

$$\therefore S = 27 + (19 + 19) + (11 + 11) + 3 = 90 \text{ cm}$$

Section – C

14. 02.00

15. 01.00

Sol. (for Q.14-15):

$$\frac{2mv_0}{qB_0} = \frac{1}{2} \left(\frac{qE_0}{m} \right) t^2$$

$$\frac{4m^2 v_0}{q^2 E_0 B_0} = t^2$$

$$x = v_0 \times t \Rightarrow x = v_0 \frac{2m}{q} \sqrt{\frac{v_0}{E_0 B_0}}$$

16. 01.00

17. 01.00

Sol. (for Q. 16-17):

For solid sphere to just start rolling without slipping.

$$\mu = \frac{\tan \theta}{1 + \frac{mR^2}{I}} = \frac{2}{7} \tan \theta$$

$$a_{\text{sphere}} = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{5}{7} g \sin \theta$$

$$a_{\text{cylinder}} = g(\sin \theta - \mu \cos \theta) = \frac{5}{7} g \sin \theta$$

$$a_{\text{ring}} = g(\sin \theta - \mu \cos \theta) = \frac{5}{7} g \sin \theta$$

Ring, solid cylinder and solid sphere have same acceleration.

18. 01.00

19. 03.00

Sol. **(for Q. 18-19):**

During the first crossing,

$$\frac{d}{v} = t \Rightarrow v = \frac{d}{t}$$

$$\Rightarrow X = 1$$

During the second crossing,

$$\sin \theta = \frac{v}{u} \text{ and } \frac{d}{v \cos \theta} = 3t \Rightarrow \cos \theta = 1/3$$

$$u = \frac{v}{\sin \theta} = \frac{3}{2\sqrt{2}} \frac{d}{t}$$

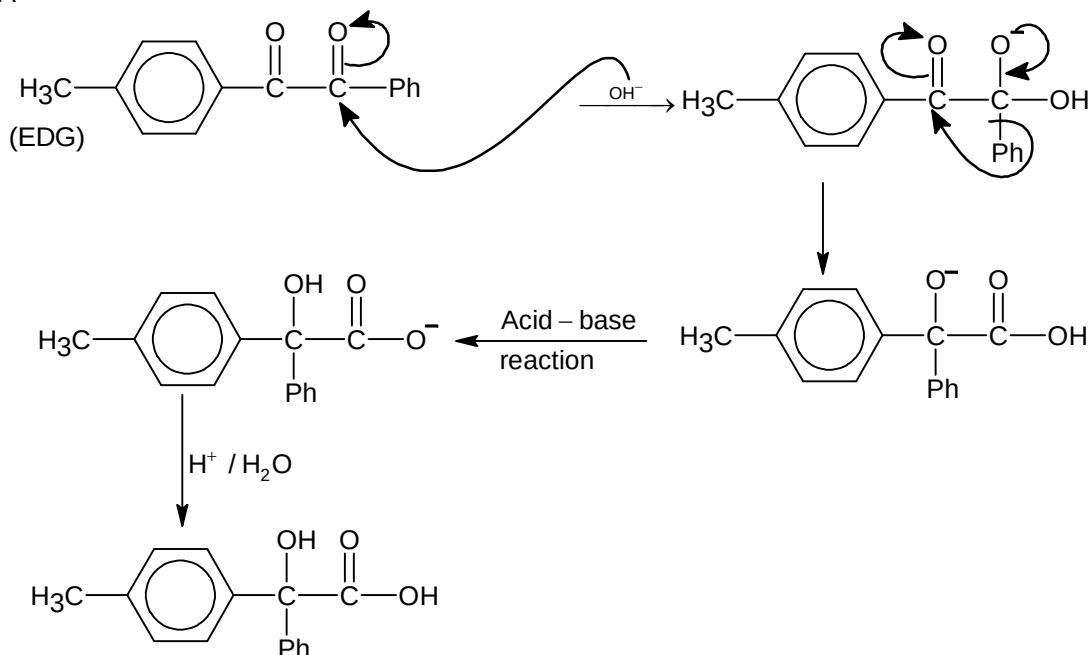
$$\Rightarrow Y = 3$$

Chemistry

PART – II

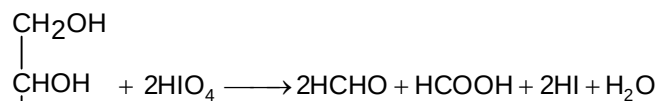
Section – A

20. A
Sol.

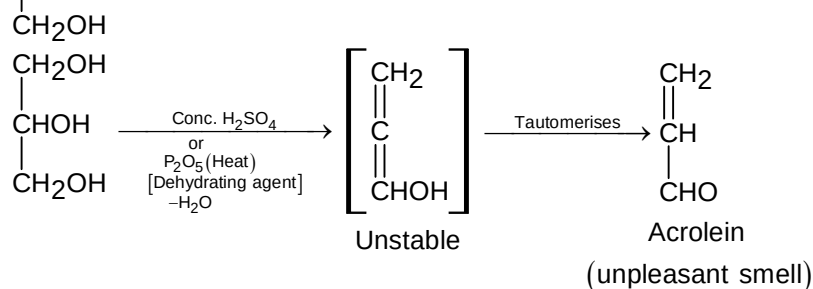


21. B
Sol. Beryl is cyclic silicate, two oxygen atoms per tetrahedron are shared to form closed ring.

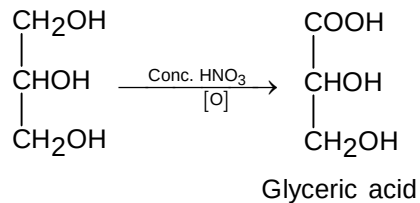
22. D
Sol. (A)

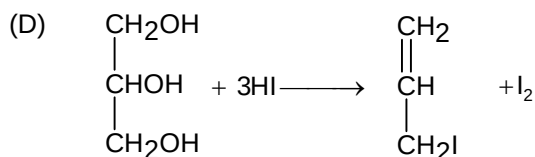


(B)



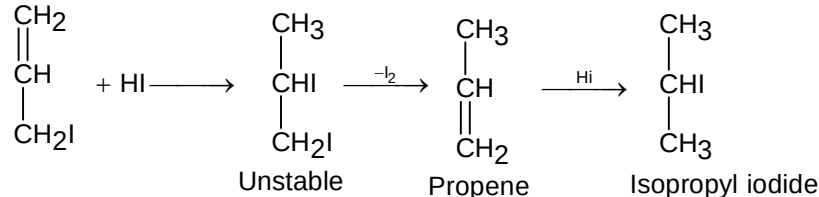
(C)





Allyl iodide

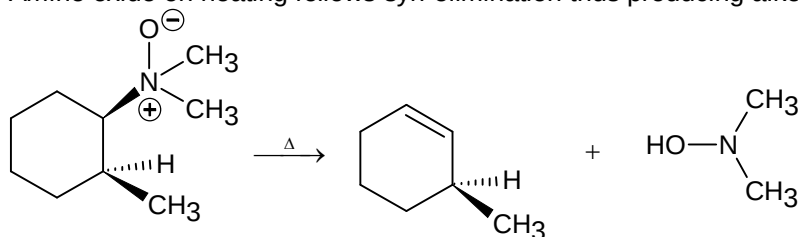
Allyl iodide formed further reacts with HI.



23.

B

Sol. Amine oxide on heating follows syn-elimination thus producing alkene



24.

B, C, D

Sol.
$$E_{\text{cell}} = \frac{0.059}{2} \log \frac{[\text{M}^{2+}]_{\text{right}}}{[\text{M}^{2+}]_{\text{left}}}$$

$$0.236 = \frac{0.059}{2} \log \frac{[0.01]}{[\text{M}^{2+}]_{\text{left}}}$$

$$[\text{M}^{2+}]_{\text{left}} = 1 \times 10^{-10} \text{ M}$$

$$\therefore \text{Solubility of } \text{MX}_2 \text{ in } 0.2 \text{ M } \text{X}^- = 1 \times 10^{-10} \text{ M}$$

$$K_{\text{SP}} \text{ of } \text{MX}_2 = [\text{M}^{2+}][\text{X}^-]^2 = 1 \times 10^{-10} \times (0.2)^2 = 4 \times 10^{-12} \text{ M}^3$$

$$\text{Solubility of } \text{MX}_2 \text{ in water} = \left(\frac{K_{\text{SP}}}{4} \right)^{1/3} = \left(\frac{4.0 \times 10^{-12}}{4} \right)^{1/3} = 1.0 \times 10^{-4} \text{ M}$$

25.

A, B, C, D

Sol. Cellulose, a polymer of D-glucose, has β -1,4-glycosidic linkage.

26.

A, C

Sol.

Haematite – Fe_2O_3	Bauxite – $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$	} Oxide ores of same metal
Magnetite – Fe_3O_4	Corundum – Al_2O_3	
Anglesite – PbSO_4	Cuprite – Cu_2O	} Oxide ores of different metals
Cerrusite – PbCO_3	Cassiterite – SnO_2	

27. B, C, D

Sol. Kjeldahl's method is not applicable to compounds in which nitrogen atom is present in ring and to compounds containing nitro and azo groups.

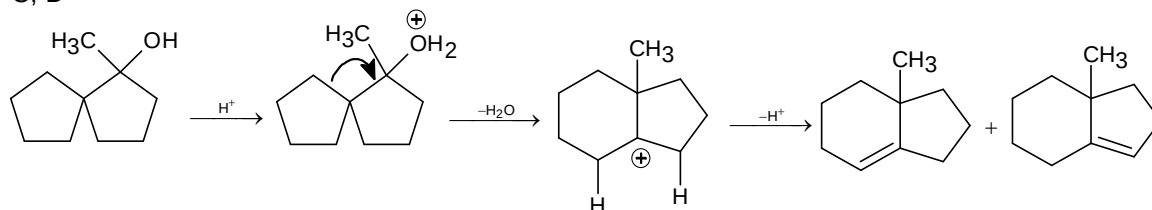
28. B, D

 Sol. Ice cube of D_2O has greater density than H_2O (liquid) and hence will sink.

 Ice cube H_2O has smaller density than D_2O (liquid) and hence will float on it.

29. C, D

Sol.



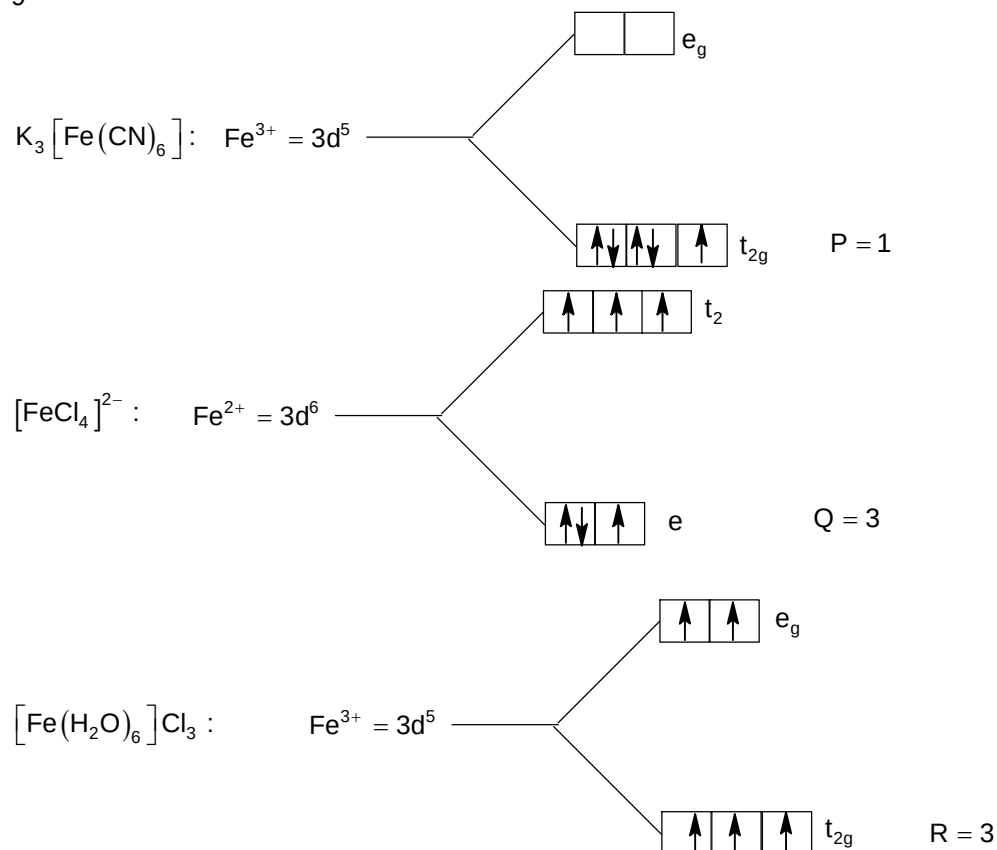
Section – B

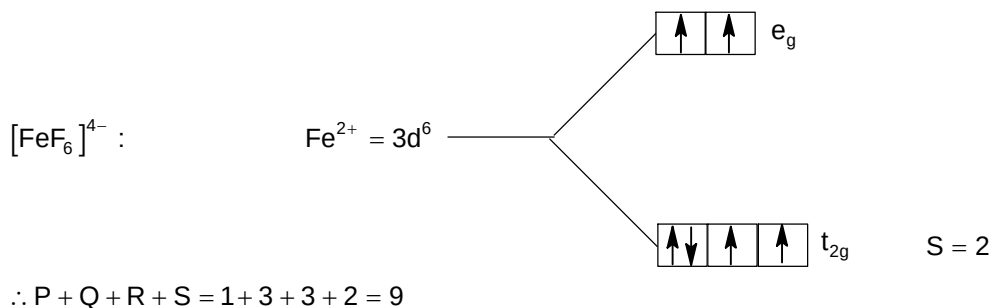
30. 7

 Sol. All statements except (F) are true.
 Insulin is a hormone and not co-enzyme.

31. 9

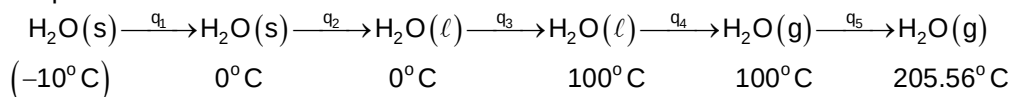
Sol.





32. 17

Sol. the process involved are



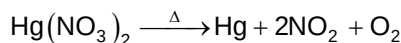
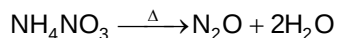
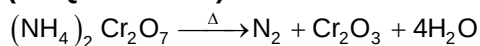
$$\begin{aligned}
 Q &= q_1 + q_2 + q_3 + q_4 + q_5 \\
 &= 5[2 \times 10 + 333 + 4.18 \times 100 + 2439 + 1.8 \times 105.56] \\
 &= \mathbf{17000 \text{ J}}
 \end{aligned}$$

Section – C

33. 44.80

34. 67.20

Sol. (for Q. 33 and 34):



$$\begin{aligned}
 \text{Total number of moles of diatomic gases (X)} &= \text{N}_2 (1 \text{ mole}) + \text{O}_2 (1 \text{ mole}) \\
 &= 2 \text{ moles}
 \end{aligned}$$

$$\therefore \text{Volume} = 2 \times 22.4 = 44.8 \text{ litres}$$

$$\begin{aligned}
 \text{Total number of moles of triatomic gases} &= \text{N}_2\text{O} (1 \text{ mole}) + \text{NO}_2 (2 \text{ moles}) \\
 &= 3 \text{ moles}
 \end{aligned}$$

$$\therefore \text{Volume} = 3 \times 22.4 = 67.2 \text{ litres}$$

35. 272.87
(272.86 – 272.88)

36. 00.09

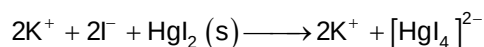
Sol. (for Q. 35 and 36):

$$\text{Molality of KI solution} = \frac{1.66 \times 1000}{166 \times 100} = 0.1$$

$$\text{Effective molality} = i \times 0.1 = 2 \times 0.1 = 0.2$$

$$\Delta T_f (\text{KI}) = K_f \times m = 1.86 \times 0.2 = 0.372$$

$$\text{'X' freezing point of KI solution} = 273.15 - 0.372 = 272.778 = 272.78 \text{ K}$$

In KI solution saturated with solid HgI_2 we get $\text{K}_2[\text{HgI}_4]$ 

$$\text{Now, effective molality} = 1.5 \times 0.1 = 0.15$$

$$\begin{aligned}
 \text{'Y' (Freezing point of saturated solution)} &= 273.15 - (1.86 \times 0.15) \\
 &= 273.15 - 0.279 \\
 &= 272.87 \text{ K}
 \end{aligned}$$

$$\begin{aligned}
 \therefore Y - X &= 272.87 - 272.78 \\
 &= 0.09 \text{ K}
 \end{aligned}$$

37. 03.00

38. 30.60

Sol. (for Q. 37 and 38):

$$\Delta E = 3.4 \text{ eV} = 13.6Z^2 \left[\frac{1}{n^2} - \frac{1}{\infty^2} \right]$$

$$3.4 = 13.6 \frac{Z^2}{n^2} \quad \dots (1)$$

Energy emitted in the transition from excited state to the IInd excited state ($n = 3$)

$$\Delta E = 4.25 + 5.95 = 13.6Z^2 \left[\frac{1}{3^2} - \frac{1}{n^2} \right]$$

$$10.20 = 13.6 Z^2 \left[\frac{n^2 - 9}{9n^2} \right] \quad \dots (2)$$

From (1) and (2), we get

$$\frac{10.20}{3.4} = \frac{n^2 - 9}{9}$$

$$\Rightarrow 3 = \frac{n^2 - 9}{9} \Rightarrow n = 6$$

$$\text{From Eq. (1)} \quad Z^2 = \frac{3.40 \times n^2}{13.6} \Rightarrow Z = 3$$

$$\text{Ionization energy of first excited state (n = 2) of the atomic species } \Delta E = 13.6 \times 3^2 \left[\frac{1}{2^2} - \frac{1}{\infty^2} \right]$$

$$= 13.6 \times 9 \times \frac{1}{4}$$

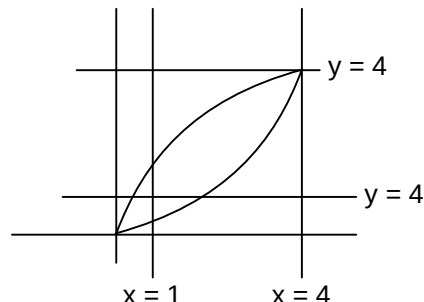
$$= 30.6 \text{ eV}$$

Mathematics**PART – III****Section – A**

39. D

Sol. $y^2 = 4[\sqrt{y}]x \Rightarrow y \in [1, 4] [\sqrt{y}] = 1$
 $\Rightarrow y^2 = 4x$ similarly $x \in [1, 4] [\sqrt{x}] = 1$
 $x^2 = 4y$

Required area $\int_1^2 (2\sqrt{x} - 1) dx + \int_2^4 \left(2\sqrt{x} - \frac{x^2}{4}\right) dx = \frac{11}{3}$



40. A

Sol. $\int_0^x (x-t+1)g(t) dt = x^4 + x^2$; differentiating with respect to x

$g(x) + \int_0^x g(t) dt = 4x^3 + 2x$; differentiating again $g'(x) + g(x) = 12x^2 + 2$

So, $g(x) = 12x^2 - 24x + 26 - 26e^{-x}$

41. A

Sol. The desired region consists of 12 boxes, whose lower left corners are integers solution of $x^2 + y^2 = 25$, i.e. $(\pm 5, 0)$, $(0, \pm 5)$, $(\pm 3, \pm 4)$, $(\pm 4, \pm 3)$

Since, the points them self are symmetric about $(0, 0)$ the boxes are symmetric about $\left(\frac{1}{2}, \frac{1}{2}\right)$,

then distance from $\left(\frac{1}{2}, \frac{1}{2}\right)$ to the fullest point on a box that lays on the axis

So, minimum values $r = \frac{\sqrt{130}}{2}$

42. B

Sol. $2010^2 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 67^2$

Total divisors $= 3 \times 3 \times 3 \times 3 = 3^4$ i.e. $(2+1)^4$ and $(1+1)^4$ of which are squares

So, required probability $= \frac{26}{81}$

43. A, C

Sol. Let the centre of base is (0)

$\overline{AB} = 2\hat{j} \Rightarrow |\overline{AB}| = 2 \Rightarrow \Delta OAB = \frac{1}{4} \cdot 4\sqrt{3}$

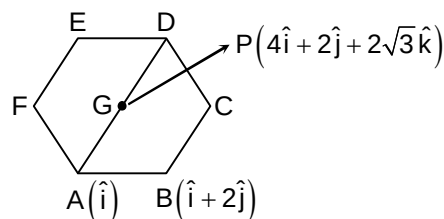
$\Rightarrow$ Base area $= 6\sqrt{3}$

Let height of pyramid $= h$

$\Rightarrow \frac{1}{3} \cdot 6\sqrt{3} h = 6\sqrt{3} \Rightarrow h = 3$

$\overline{AP} = 3\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k} \Rightarrow |\overline{AP}| = 5 \Rightarrow |\overline{AG}| = 4$. Now $\overline{AG}$ and $\overline{AO}$ are collinear

$\Rightarrow \overline{AG} = \lambda \overline{AO} \Rightarrow |\overline{AG}| = |\lambda| \cdot |\overline{AO}| \Rightarrow \lambda = \pm 2 \Rightarrow \vec{G} = \hat{i} \pm 2(\hat{j} + \sqrt{3}\hat{i})$



$$\Rightarrow \vec{G} = \hat{i} - 2\hat{j} - 2\sqrt{3}\hat{k} \text{ or } \hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$$

44. A, B, C, D

Sol. Since, $|\sqrt{1+\sin 2x} - \sqrt{1-\sin 2x}| \leq \sqrt{2} \quad \forall x \in [0, 2\pi]$

$$\text{Now, if } \cos x \leq 0 \Rightarrow 2 \cos x \leq |\sqrt{1+\sin 2x} - \sqrt{1-\sin 2x}| \Rightarrow x \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

$$\text{If } \cos x > 0, \text{ then } 4 \cos^2 x \leq 1 + \sin 2x + 1 - \sin 2x - 2\sqrt{1-\sin^2 2x} \Rightarrow \cos 2x \leq 0$$

$$x \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right] \cup \left[\frac{5\pi}{4}, \frac{7\pi}{4}\right] \Rightarrow x \in \left[\frac{\pi}{4}, \frac{7\pi}{4}\right] \text{ so, } x = 1, 2, 3, 4$$

45. B, C

Sol. $4^{\sec^2 A} x^2 + 2x + \left(B^2 - B + \frac{1}{2}\right) = 0$ have real roots

$$\Rightarrow D \geq 0 \Rightarrow 4^{\sec^2 A} \left(B^2 - B + \frac{1}{2}\right) \leq 1 \text{ but } 4^{\sec^2 A} \geq 4, B^2 - B + \frac{1}{2} \geq \frac{1}{4}$$

$$\Rightarrow \text{Equation will satisfy when } 4^{\sec^2 A} = 4 \text{ and } B^2 - B + \frac{1}{2} = \frac{1}{4}$$

$$\Rightarrow \sec^2 A = 1 \text{ and } \left(B - \frac{1}{2}\right)^2 = 0 \Rightarrow \cos^2 A = 1 \text{ and } B = \frac{1}{2}$$

$$\Rightarrow A = n\pi \text{ and } B = \frac{1}{2} \Rightarrow \cos A + \cos^{-1} B = \frac{\pi}{3} - 1; \frac{\pi}{3} + 1$$

46. A, C

Sol. Let $\vec{p}$ and $\vec{q}$ be the pv of P and Q

$$\text{Let } \vec{p} = \vec{a} + t_1 \vec{\alpha}; \vec{q} = \vec{b} + s_1 \vec{\beta}$$

$$\Rightarrow \vec{p} - \vec{q} = \vec{a} - \vec{b} + t_1 \vec{\alpha} - s_1 \vec{\beta}$$

Taking dot product of this equation with $\vec{\alpha}$ and $\vec{\beta}$, we have

$$0 = (\vec{a} - \vec{b}) \cdot \vec{\alpha} + t_1 (\vec{\alpha} \cdot \vec{\alpha}) - s_1 (\vec{\alpha} \cdot \vec{\beta}) \text{ and}$$

$$0 = (\vec{a} - \vec{b}) \cdot \vec{\beta} + t_1 (\vec{\alpha} \cdot \vec{\beta}) - s_1 (\vec{\beta} \cdot \vec{\beta}) \text{ solving for } t_1 \text{ and } t_2, \text{ we get } \vec{p} \text{ and } \vec{q}$$

$$\begin{array}{c} \vec{r} = \vec{a} + \lambda \vec{\alpha} \\ \text{P} \qquad \qquad \text{A}(\vec{a}) \\ \vec{r} = \vec{b} + \mu \vec{\beta} \\ \text{Q} \qquad \qquad \text{B}(\vec{b}) \end{array}$$

47. A, C

Sol. Since, $x^3 + ax + b = 0$; $x^3 + cx^2 + d = 0$ have complex root so it appears in conjugate pairs

$$\text{Let } z = m + \sqrt{n}i; \bar{z} = m - \sqrt{n}i; \text{ So } z + \bar{z} = 2m; z\bar{z} = m^2 + n$$

$$\text{For the equation } x^3 + ax + b = 0 \Rightarrow -20 + z + \bar{z} = 0; z + \bar{z} = 20 \Rightarrow m = 10$$

$$\text{and for equation } x^3 + cx^2 + d = 0 \Rightarrow -21z - 21\bar{z} + z\bar{z} = 0 \Rightarrow n = 320$$

48. C, D

Sol. Clearly $f(x)$ is not define at $x = 1$ if $a < 0$ and if $a > 0$, then all the conditions of Rolle's theorem holds on $f(x)$

Section - B

49. 1

Sol. Since, $g(0) = 0$ and $f(g(x)) = x$

$$\Rightarrow g'''(x) = 3(f'(g(x)))^{-5} (f''(g(x)))^2 - (f'(g(x)))^{-4} \cdot f'''(g(x)) \Rightarrow g'''(0) = 1$$

50. 13

Sol. E, F exist if and only if $\frac{\text{Area of } \triangle ADC}{\text{Area of } \triangle ADB} = 2$

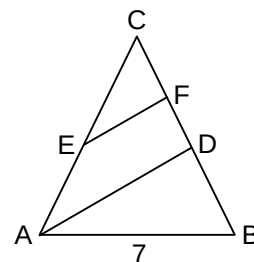
Since, AD is the angle bisector, and the ratio of area of triangles with equal height is the ratio of their bases

$$\frac{AC}{AB} = \frac{DC}{DB} = \frac{\text{Area ADC}}{\text{Area ADB}} = 2 \Rightarrow AC = 2AB$$

Let BC = d so, by triangle inequality $d + 7 > 14$; $7 + 14 > d$

$$\Rightarrow 7 < d < 21$$

So, there are 13 possible integral values of BC



51. 5

Sol. Put $t = \frac{1}{x} \Rightarrow \lim_{t \rightarrow 0} t^{-p} \left(\left(\frac{1}{t} + 1 \right)^{\frac{1}{3}} + \left(\frac{1}{t} - 1 \right)^{\frac{1}{3}} - 2 \left(\frac{1}{t} \right)^{\frac{1}{3}} \right)$

$$\Rightarrow \lim_{t \rightarrow 0} t^{-p-\frac{1}{3}} \left((1+t)^{\frac{1}{3}} + (1-t)^{\frac{1}{3}} - 2 \right) \text{ use expansion } (1+t)^{\frac{1}{3}} = 1 + \frac{t}{3} - \frac{t^2}{9}; (1-t)^{\frac{1}{3}} = 1 - \frac{t}{3} - \frac{t^2}{9}$$

$$\Rightarrow \lim_{t \rightarrow 0} t^{-p-\frac{1}{3}} \left(-\frac{2t^2}{9} \right) \Rightarrow -\frac{2}{9} \cdot t^{-p+2-\frac{1}{3}} = -\frac{2}{9} \cdot t^{-p+\frac{5}{3}} \text{ so, } p = \frac{5}{3} \Rightarrow 3p = 5$$

Section - C

52. 04.00

53. 06.00

Sol. (for Q. 52 to 53):

$$f(x+2y) = f(x) + f(2y) + e^{x+2y}(x+2y) - x e^x - 2y e^{2y} + 4xy$$

Replace $x, y = 0 \Rightarrow f(0) = 0$

$$\text{Put } 2y = -x, \text{ we have } f(0) = f(x) + f(-x) - x e^x + x e^{-x} - 2x^2$$

$$\Rightarrow -f(x) = f(-x) - x e^x + x e^{-x} - 2x^2. \text{ Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) + f(-x) - x e^x + x e^{-x} - 2x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - e^h h + (x+h) e^{x+h} - x e^{-x} + 2(x+h)x - x e^x + x e^{-x} - 2x^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{f(h) - e^h h + x e^x e^h + 2hx - x e^x}{h} = \lim_{h \rightarrow 0} \frac{(f(h) - f(0)) - e^h h (e^x - 1) + x e^x (e^h - 1) + 2hx}{h}$$

$$\Rightarrow f(x) = x^2 + x e^x \Rightarrow f(2) = 4 + 2e^2$$

54. 84.00

55. 25.00

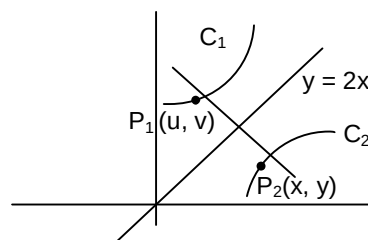
Sol. (for Q. 54 to 55):

Let $P_1(u, v)$ be any point of C_1 and $P_2(x, y)$ be the corresponding point on C_2

Since, $P_1 P_2$ (line joining) is perpendicular to $y = 2x$ so,

$$\text{its slope} = -\frac{1}{2}$$

$$\frac{y-v}{2x-u} = -\frac{1}{2} \quad \dots (1)$$



and midpoint of P_1P_2 lies on $y = 2x$

$$\frac{y+v}{2} = 2\left(\frac{x+u}{2}\right) \quad \dots (2)$$

Solving equation (1) and (2), we get $12x^2 - 7xy - 12y^2 + 25 = 0$

56. 401.00

57. 10.00

Sol. **(for Q. 56 to 57):**

$$f(2+x) = f(2-x) \Rightarrow f(4-x) = f(x) \quad \dots (1)$$

$$\text{From } f(7+x) = f(7-x) \Rightarrow f(14-x) = f(x) \quad \dots (2)$$

From equation (1) and (2), we get $f(x+10) = f(x)$ so period is 10

Now, replacing x by $x+10$ and then $x-10$ is equation (3) continuity in this ways

$$f(x+10n) = f(x) \quad \dots (4)$$

for $n = \pm 1, \pm 2, \pm 3 \dots$

Since, $f(0) = 0$ equation (4)

$$f(\pm 10) = f(\pm 20) \dots = f(\pm 1000) = 0. \text{ so total } 201$$

Now, $x = 0$, in equation (1) $f(x) = f(0) = 0$ from equation (1), we have 200 more roots